National University

of Computer & Emerging Sciences

MT-1006 Differential Equations

Assignment No:03 **Individual Assignment**

Semester: Fall 2023 **Marks: 10 × 10**

Due date: Monday, December 04, 2023. **Sections: BCS, BAI**

1. Plagiarized work will result in zero marks.
2. No retake or late submission will be accepted.
3. Submit the assignment before the deadline.
4. You must submit the assignment on the proper sheets, don't use papers from the rough register.

Question 1: For the following questions, just write down the trial particular solution y_p from the given input function $g(x)$.

i. $y'' - 8y' + y = 100x^2 - 26x\, e^x$

ii. $y'' + 4y = (x^2 - 3)\sin 2x$

iii. $y'' - 2y' + 5y = e^x \cos 2x$

iv. $y'' - 2y' + 2y = e^{2x}(\cos 2x + 3\sin x)$

v. $y''' - y'' - 4y' + 4y = 5 - e^x + e^{2x}$

vi. $y'' - 4y' + 8y = (10x^2 - x - 1)e^{2x}\sin 2x$

Question 2: solve the given differential equation by undetermined coefficients.

$$y'' + 6y' + 9y = 6x^2 + 2 - 12e^{3x} - \sin 2x$$

Question 3: Find the general solution of the given non-homogeneous equation**.**

a. $x^2y'' - xy' + y = \ln x$

b. $x^2y'' + xy' + y = \sec(\ln x)$

c. $x^3y''' - 3x^2y'' + 6xy' - 6y = 3 + \ln x^3$

Question 4: Solve the initial value problem.

a. $y'' - y' + y = \frac{e^t}{t^2};\ y(0) = 0; y'(0) = 0$

b. $y'' + y = 3\sin^2 x\ ; y(0) = 0; y'(0) = 0$

c. $y''' - 3y'' + 2y' = \frac{e^{2x}}{1+e^x}; y(0) = 1; y'(0) = 0$